Hybrid Event-Driven Spoofing Detection (HEDSD) method Wireless Sensor Networks

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***Abstract*—This paper proposes an energy-adaptive hybrid spoofing detection framework for resource-constrained industrial Wireless Sensor Networks (WSNs), addressing the fundamental trade-off between detection accuracy and sensor battery longevity. While continuous public-key cryptographic authentication (ECDSA NIST P-256) guarantees unforgeable origin security, executing signature verification on every packet rapidly exhausts node battery reserves (90.0 mJ per check). Conversely, lightweight physical-layer fingerprinting schemes (5 μJ) suffer elevated false alarms under industrial channel fading. To resolve this dilemma, the proposed framework introduces a multi-tier decision architecture combining physical-layer spatial RSSI screening, an event-driven trigger mechanism, context-aware dynamic trust scoring ($T = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$), and an Early Warning Score (EWS) temporal accumulator. Unlike existing approaches relying on continuous monitoring or heavy computational models, our framework selectively activates cryptographic signature audits strictly upon anomaly detection while employing a three-tier energy-adaptive mode switching strategy that gracefully degrades evaluation complexity as battery levels decline. Validated through discrete-event network simulations across a 401-node industrial factory floor processing over 1.1 million telemetry packets, the proposed model selectively triggers heavy cryptographic audits on only 0.53% of incoming traffic, achieving a 99.40% reduction in per-packet evaluation energy (from 90.00 mJ down to 0.540 mJ). The framework delivers a 98.18% detection accuracy with an exceptionally low 0.03% false positive rate, preserving 63.4% of cluster head battery capacity and extending network operational lifetime by over 130-fold.**

***Index Terms*—Wireless Sensor Networks, Identity Spoofing Detection, Physical-Layer Security, Adaptive Trust Management, Resource Minimization, Industrial Internet of Things.**

I. Introduction

A spoofing attack in wireless communication networks is a critical security threat where an unauthorized adversary maliciously assumes the identity of a valid device by forging its network credentials or physical-layer transmission signatures [7]. Through deception as a genuine node, the adversary avoids authorization at the periphery, accesses the broadcast wireless channel shared by the network, and communicates unauthorizedly with other nodes or gateways. As a result, the deceiver is able to inject fake measurements, alter the routing decisions made based on those measurements, cause disruption through denial-of-service attacks, and violate the confidentiality and integrity of the entire network [6]. Because of the openness of radio waves and the lack of any physical protection for low-power wireless devices, identity spoofing has become one of the most widespread vulnerabilities of modern wireless technologies.

Wireless Sensor Networks (WSNs) constitute the digital infrastructure of the Industrial Internet of Things (IIoT) and facilitate the continuous monitoring of the environment, automation, control, and predictive maintenance in smart factories and critical infrastructures [1], [2]. In the typical industrial deployment, hundreds of low-power wireless sensor nodes continuously collect the telemetry data (e.g., temperature, vibrations, pressure, and flow rates) and forward packets in multi-hop clustered architectures to a central Base Station (BS) or industrial edge gateway [3], [4].

However, because of the open-broadcast characteristic of wireless channels for communication, no physical tamper resistance in industrial field devices, and the requirement for minimal energy consumption, industrial WSNs become highly susceptible to wireless attacks [5], [6]. Identity Spoofing attacks and **Sybil Attacks** represent such security threats. In an identity spoofing attack, the attacker pretends to be a genuine sensor by imitating its MAC address or network layer ID, introducing malicious control signals, corrupting sensor readings, and disrupting routing of the cluster [7], [8]. Sybil attacks involve one rogue sensor node claiming different fake sensor IDs at the same time, making it dominant in the cluster head election process and data aggregation procedure [9].

To protect against origin spoofing, conventional network security frameworks mandate continuous public-key cryptographic signature verification, most notably using Elliptic Curve Cryptography (ECDSA NIST P-256) or HMAC authentication for every transmitted packet [10], [11]. Although continuous cryptographic authentication delivers robust security assurances against identity forgery, it

incurs an unsustainable computational and energy tax on resource-constrained sensor hardware [12]. Experimental benchmarks on 16-bit MSP430-class sensor microcontrollers (operating at 8 MHz) equipped with CC2420 transceivers demonstrate that verifying a single ECDSA-P256 digital signature consumes approximately 90.0 mJ of electrical energy [43] and takes upwards of 1.2 seconds of processing time [13], [14]. For Cluster Heads (CHs) processing thousands of telemetry packets per hour, continuous signature verification leads to rapid battery depletion, node premature failure, network partitioning, and catastrophic loss of operational monitoring capacity [15].
Conversely, lightweight non-cryptographic anomaly detection schemes (such as pure Received Signal Strength Indicator (RSSI) fingerprinting or standalone machine learning classifiers) offer micro-joule operational overhead (5–100 μJ), but suffer from elevated False Positive Rates (FPR) under industrial radio interference, link quality fluctuations [29], multipath fading, and dynamic shadowing effects [45]. Pure ML models operating in static regimes fail to adapt when physical channel conditions fluctuate or when network energy levels drop, leading to either security breaches or unnecessary alarm generation [18].
To resolve this critical energy-security trade-off, this paper proposes the **Hybrid Event-Driven Spoofing Detection (HEDSD)** framework. The proposed model adopts an industrial adaptive decision method, bridging physical-layer radio propagation properties, lightweight machine learning, dynamic multi-factor trust evaluation, and conditional cryptographic verification into a multi-tier defense architecture tailored for large-scale industrial WSNs.

A. Research Motivations and Key Contributions

The key driver for the work presented here comes from the challenging resource-security trade-off in industrial WSNs, where while continuous public key cryptography-based authentication ensures tamper-proof origination, it drains node battery power significantly (90.0 mJ per authentication), whereas while lightweight physical layer filtering (5 J per check) has a high false alarm rate. To overcome this trade-off, this paper introduces a multi-tier hybrid spoofing detection framework that selectively restricts public-key signature verification to verified anomaly events while employing energy-adaptive mode switching as battery reserves decline. The key contributions of this work include: (i) a multi-tier adaptive architecture that integrates physical-layer RSSI screening with conditional ECDSA-P256 signature auditing to minimize steady-state energy consumption; (ii) a context-aware dynamic trust model ($T_i = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$) featuring threat-driven weight adaptation (α, β, γ) to dynamically balance cryptographic reliance against battery conservation; (iii) a temporal Early Warning Score accumulator (EWS_i) and joint isolation rule to suppress transient fading spikes and eliminate false quarantining (0.03% FPR); and (iv) extensive discrete-event simulation across a 401-node industrial network validating a 99.40% reduction in per-packet evaluation energy (0.540 mJ/pkt), 98.18% spoofing detection accuracy, and a 130-fold operational lifetime extension.

II. Related Work and Literature Survey

Spoofing attack detection in Wireless Sensor Networks has been widely studied across four primary paradigms: (i) Physical-Layer Radio Fingerprinting, (ii) Cryptographic Origin Authentication, (iii) Trust and Reputation Security Models, and (iv) Machine Learning Anomaly Classifiers.

A. Physical-Layer Radio Fingerprinting

Physical-layer security approaches leverage inherent physical characteristics of wireless radio propagation, such as Received Signal Strength Indicator (RSSI), Channel State Information (CSI), Phase Difference of Arrival (PDoA), and Angle of Arrival (AoA), to authenticate node origin without relying on secret keys [5], [19].
Yang *et al.* [20] proposed K-means clustering over RSSI spatial profiles to detect identity spoofing in IEEE 802.11 networks. Their approach demonstrated that spatial correlation between RSSI readings from different antenna receivers can effectively distinguish legitimate transmissions from spoofed signals originating from distinct spatial locations. However, their model assumed stationary radio channels without heavy industrial multipath fading.
Liu *et al.* [21] utilized fine-grained CSI amplitude and phase features from OFDM subcarriers to detect spoofing in industrial wireless networks. While CSI provides higher spatial granularity than RSSI, acquiring CSI requires specialized software-defined radio (SDR) hardware or modified Wi-Fi NIC drivers, rendering it impractical for low-cost IEEE 802.15.4 / ZigBee sensor nodes. Furthermore, environmental dynamics such as moving industrial machinery or overhead cranes cause rapid spatial decorrelation, resulting in elevated false positive rates ($> 12\%$) [22].

B. Cryptographic Origin Authentication Schemes

Cryptographic approaches guarantee origin authenticity using symmetric MACs or asymmetric digital signatures [11]. Heinzelman *et al.* [23] integrated symmetric key pre-distribution into clustered WSN architectures (SecLEACH). While symmetric HMAC checks incur low computational overhead ($< 10\,\mu$J), key compromise at a single sensor node exposes the entire cluster to impersonation and Sybil attacks.
To prevent master key compromise, public-key cryptographic primitives, specifically Elliptic Curve Cryptography (ECC), have been implemented for WSNs [10]. Public-key schemes like TinyECC [24] and ECDSA NIST P-256 enable unforgeable per-packet signature generation and verification. However, performance studies by Piotrowski *et al.* [12] and Wandeler *et al.* [14] proved that verifying an ECDSA-P256 signature on 16-bit sensor microcontrollers

(e.g., TI MSP430 or Atmel ATmega128L) requires 90.0–120.0 mJ per packet. In dense industrial telemetry scenarios where cluster heads receive thousands of packets per hour, continuous cryptographic verification causes rapid battery drain, shortening cluster head lifetime to a few days [15].

C. Trust and Reputation Management Systems

Trust-based security systems evaluate node trustworthiness by monitoring historical communication behaviors, packet forwarding ratios, and sequence number continuity [18], [25]. Ganeriwal *et al.* [26] introduced the Reputation-based Framework for Sensor Networks (RFSN), utilizing Beta reputation distributions to compute trust scores.

Sun *et al.* [27] developed a multi-attribute dynamic trust model incorporating energy consumption and packet delivery metrics. Similarly, Zhang *et al.* [28] applied fuzzy trust logic to isolate Sybil nodes in multi-hop WSNs. Recently, Wang *et al.* [46] (IEEE TMC 2024) proposed a dynamic trust management scheme for malicious node detection in weak-link WSNs, utilizing adaptive behavior observation. However, existing dynamic trust models process every transmitted frame continuously, incurring constant computational overhead regardless of node battery drain or network threat level.

D. Machine Learning-Assisted Anomaly Classifiers

Recent advances in machine learning (ML) have led to the adoption of supervised and unsupervised anomaly classifiers for WSN intrusion detection [17], [30], [50]. Machine learning algorithms, including Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), and XGBoost, extract multi-dimensional features (such as packet arrival intervals, RSSI variations, and sequence gaps) to detect spoofing patterns [31], [32].

Zhang *et al.* [47] (IEEE Sensors J. 2022) developed a multi-parameter spoofing detection scheme leveraging Support Vector Machines (SVM) on physical signal parameters. Tan *et al.* [33] and Vinayakumar *et al.* [34] implemented deep neural network (DNN) architectures for intrusion and spoofing detection in IoT networks, achieving 98.5% detection accuracy. Al-Fuhaidi *et al.* [50] evaluated anomaly-based WSN intrusion detection using ensemble classifiers. However, running continuous ML inference on memory-constrained sensor microcontrollers ($<$ 10 KB RAM) is computationally expensive (100–$12,500\,\mu$J per packet). Standalone ML classifiers operating without physical spatial verification or selective cryptographic fallback remain vulnerable to adversarial evasion attacks and mimicry spoofing [35].

E. Hybrid Trust-ML and Physical-Layer Authentication

To combine physical signal features with trust logic, Ali *et al.* [48] proposed a multi-attribute physical-layer authentication (PLA) framework utilizing RSSI, battery levels, and machine learning classifiers. Kumar *et al.* [49] (IEEE IoTJ 2024) introduced a trust-aware authentication scheme combining federated machine learning with behavioral trust scoring. While federated trust models deliver strong security assurances, their high communication overhead and intensive model aggregation cycles make them unsuitable for low-power IEEE 802.15.4 sensor nodes.

F. Literature Summary and Research Gaps

Existing WSN spoofing detection approaches—spanning physical RSSI/CSI profiling [20], [21], continuous ECDSA authentication [12], dynamic trust systems [46], SVM classifiers [47], multi-attribute PLA [48], federated learning [49], and ensemble IDS [50]—are indexed across top-tier computer science venues. As summarized in Table I, current techniques either exhibit high sensitivity to industrial channel fading or incur prohibitive computational costs (0.065–90.00 mJ/pkt) without providing multi-tier adaptive trigger capabilities.

III. System Model and Industrial Scenario Setup

We model an industrial factory floor (1000 m $\times$ 1000 m) operating a two-tier clustered Wireless Sensor Network of $N = 401$ nodes, comprising a central line-powered Base Station, 40 Cluster Heads ($E_{\mathrm{CH}}^{(0)} = 50.0$ J), 359 leaf nodes ($E_{\mathrm{leaf}}^{(0)} = 5.0$ J), and 60 active attacker nodes (15% density). Sensor energy depletion is driven by packet transceiving ($13.36\,\mu$J/bit) and algorithmic security execution across three processing tiers: Tier 1 scalar RSSI screening ($5.0\,\mu$J/pkt), Tier 2 Decision Tree ML inference ($100.0\,\mu$J/pkt), and Tier 3 ECDSA NIST P-256 signature verification (90.0 mJ per audit) [12], [14]. Cumulative node energy depletion is tracked by aggregating these transceiving and execution costs per packet.

Radio propagation experiences distance attenuation and industrial log-normal shadowing noise. The baseline Received Signal Strength Indicator (RSSI_{ij}) for a packet transmitted from physical coordinates $\mathbf{p}_i$ to Cluster Head j at distance $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ is governed by:

$$\mathrm{RSSI}_{ij} = P_{\mathrm{tx}} - 10\,n\,\log_{10}\left(\frac{d_{ij}}{d_0}\right) + X_\sigma \quad [\mathrm{dBm}] \qquad (1)$$

where transmit power $P_{\mathrm{tx}} = 0.0$ dBm, reference distance $d_0 = 1.0$ m, path loss exponent $n = 2.7$ [16], [44], under the IEEE 802.15.4 standard specification [51], and shadowing $X_\sigma \sim \mathcal{N}(0, 2.0^2\ \mathrm{dB})$. In our model, RSSI acts as a lightweight spatial consistency indicator to detect origin anomalies. We evaluate an adversary launching Intermittent Identity Spoofing (5–10 s packet bursts) and Sybil Impersonation, where attackers lack valid private ECDSA keys and inject unauthenticated or corrupted telemetry payloads.

IV. Methodology

The proposed Hybrid Event-Driven Spoofing Detection (HEDSD) framework provides a resource-efficient security

TABLE I
COMPARATIVE TAXONOMY OF STATE-OF-THE-ART WSN SPOOFING DETECTION APPROACHES VS. PROPOSED HEDSD

Detection Technique	**Overhead (mJ/Pkt)**	**Adaptive Trigger**	**Primary Technical Limitation**
Physical RSSI Clustering	0.005 mJ	No	Vulnerable to Channel Fading
Physical CSI Profiling	0.080 mJ	No	Requires Specialized SDR Hardware
Continuous ECDSA-P256	90.000 mJ	No	Severe Per-Packet Battery Drain
Dynamic Trust Management	0.065 mJ	No	Lacks Adaptive Mode Switching
SVM Multiparameter Spoofing	0.450 mJ	No	High Matrix Computation Overhead
Multi-Attribute Physical Auth	0.120 mJ	No	Lacks Multi-Factor Trust Engine
Federated Trust-ML Auth	2.500 mJ	No	Heavy Federated Parameter Sync
Anomaly WSN IDS	0.350 mJ	No	Lacks Physical Spatial Proof
Proposed HEDSD (This Work)	**0.540 mJ**	**Yes (3-Tier)**	**Optimal Security-Efficiency Balance**

decision model. To tailor the decision engine for real-world IIoT deployments, an industrial adaptive methodology is employed that combines lightweight physical indicators, dynamic trust scoring, energy-adaptive mode switching, and targeted cryptographic verification without incurring unnecessary computational tax. Wireless radio propagation in industrial environments follows predictable spatial attenuation. Because an adversary masquerading as a legitimate sensor node transmits from a different physical location, the received signal strength at the Cluster Head deviates from the expected topological baseline. The spatial RSSI anomaly metric $\Delta\text{RSSI}_i(t)$ measures this discrepancy:

$$\Delta\text{RSSI}_i(t) = \left|\text{RSSI}_{\text{obs}}^{(i,j)}(t) - \overline{\text{RSSI}}_{\text{exp}}^{(i,j)}\right| \quad [\text{dBm}] \tag{2}$$

where $\overline{\text{RSSI}}_{\text{exp}}^{(i,j)}$ is the expected baseline signal strength calculated from the spatial distance model in (1). When an impersonator transmits from an unapproved position, the spatial mismatch exceeds 5.0 dBm with high confidence ($P > 0.985$), serving as an immediate physical-layer proof of origin forgery without running expensive algorithms.

Relying on a single security metric (such as cryptographic keys or RSSI alone) creates vulnerabilities to key theft or channel noise. Our framework synthesizes three complementary security dimensions, namely cryptographic authentication (T_{auth}), physical/behavioral consistency (T_{behavior}), and node battery health (T_{energy}), into a normalized trust score $T_i(t) \in [0.0, 1.0]$:

$$T_i(t) = \alpha(t)T_{\text{auth}}^{(i)}(t) + \beta(t)T_{\text{behavior}}^{(i)}(t) + \gamma(t)T_{\text{energy}}^{(i)}(t) \tag{3}$$

subject to the constraint $\alpha(t) + \beta(t) + \gamma(t) = 1.0$. Here, T_{auth} reflects cryptographic verification outcomes (1.0 if valid, 0.0 if invalid, or cached state if bypassed); T_{behavior} penalizes signal deviations, packet rate spikes, and sequence jumps; and T_{energy} evaluates remaining node battery reserves.

Static trust weights fail when threat levels fluctuate in dynamic environments. Under normal steady-state operations, computationally expensive cryptographic audits are bypassed to conserve battery energy. When suspicious traffic patterns or signal anomalies occur, the trust engine dynamically shifts its evaluation focus. Specifically, the weight assigned to cryptographic verification (α) increases proportionally with the Early Warning Score (EWS_i), while the behavioral weight (β) adjusts according to signal variance, and the energy weight (γ) ensures overall normalization:

$$\alpha(t) = \alpha_0 + \kappa \cdot \text{EWS}_i(t), \quad \beta(t) = \beta_0 \left(1.0 - \frac{\sigma^2_{\text{RSSI}}}{\sigma^2_{\max}}\right) \tag{4}$$
$$\gamma(t) = 1.0 - (\alpha(t) + \beta(t))$$

where $\alpha_0 = 0.40, \beta_0 = 0.40, \gamma_0 = 0.20$ denote the initial baseline coefficients, while κ denotes the sensitivity scaling factor used to boost the cryptographic trust during threat states.

In order to prevent rapid depletion of the Cluster Head batteries during continuous network surveillance operations, the proposed algorithm implements an adaptive mode selection mechanism depending on the remaining ratio of battery energy ($E_{\text{ratio}} = E_{\text{rem}}/E_{\text{initial}}$). Instead of conducting heavy but static trust assessment processes continuously, the system switches between three operation modes depending on the battery power level. In the FULL Mode ($E_{\text{ratio}} > 0.70$), which is used when the battery is fully charged, the Cluster Head performs all multi-factor trust assessments and ECDSA signature audits, spending 0.540 mJ of energy on each packet evaluation process. In the LIGHT Mode ($0.30 < E_{\text{ratio}} \leq 0.70$), which is entered when the battery power level is below 70%, heavy cryptographic audits and machine learning inference are deactivated in order to save energy. In EVENT_ONLY Mode ($E_{\text{ratio}} \leq 0.30$), initiated during low power states ($\leq 30\%$), sampling is limited to every third packet, thereby lowering packet energy consumption to 0.0016 mJ for maximum node life expectancy under power constraints. This text-driven mode switching guarantees that all vital security functions continue in operation under sufficient power availability and reduces execution complexity at low power levels to ensure continued network monitoring.

In order to minimize the possibility of false alarms caused by transient noise on the communication channel, all suspicious behavior is monitored using a sliding window size of $W = 50$ packets, using the Early Warning Score ($\text{EWS}_i(t) \in [0.0, 1.0]$). A node is declared spoofed and isolated from the network if and only if three joint conditions hold simultaneously:

$$\text{Isolate}_i(t) = \mathbb{I}\left(T_i(t) < 0.60 \wedge \text{EWS}_i(t) > 0.40 \wedge \Delta\text{RSSI}_i(t) > 5.0\right) \tag{5}$$

This multi-condition rule ensures that single-packet noise spikes do not cause accidental node quarantine. Algorithm 1 summarizes the overall algorithmic procedure, verifying steady-state packets with minimal overhead ($5\,\mu$J) while invoking heavy signature verification (90 mJ) on fewer than 0.53% of total frames.

Algorithm 1: Operational Security Decision Protocol

Input: Packet p_k, Node ID i, Signal RSSI_{obs}, Battery Ratio E_{ratio}
Output: Decision $\text{Isolate}_i \in \{0, 1\}$, Updated Trust T_i

1) **if** $E_{\text{ratio}} > 0.70$ **then** Mode $\leftarrow$ FULL
   else if $E_{\text{ratio}} > 0.30$ **then** Mode $\leftarrow$ LIGHT
   else Mode $\leftarrow$ EVENT_ONLY (sample 1-in-3 pkts)
2) **set** $\Delta\text{RSSI}_i \leftarrow |\text{RSSI}_{\text{obs}} - \overline{\text{RSSI}}_{\text{exp}}|$
3) **update** dynamic weights: $\alpha \leftarrow \alpha_0 + \kappa \cdot \text{EWS}_i$, $\beta \leftarrow \beta(\sigma^2)$, $\gamma \leftarrow 1 - (\alpha + \beta)$
   compute composite trust: $T_i \leftarrow \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$
4) **if** Mode = FULL $\wedge$ $(\Delta\text{RSSI}_i > 5.0 \vee T_i < 0.60)$ **then**
   $T_{\text{auth}} \leftarrow \text{VerifyECDSA}(p_k.\text{sig})$ *// Selective Audit*
5) **if** $T_i < 0.60 \wedge \text{EWS}_i > 0.40 \wedge \Delta\text{RSSI}_i > 5.0$ **then**
   $\text{Isolate}_i \leftarrow 1$ *// Quarantine Spoofer*
   else $\text{Isolate}_i \leftarrow 0$ *// Accept Telemetry Packet*
6) **return** $(\text{Isolate}_i, T_i)$

Fig. 1. Operational pseudocode of the proposed detection protocol.

V. Experimental Setup

To evaluate the proposed security framework under realistic operational conditions, a large-scale industrial sensor network simulation was constructed across a 1000 m × 1000 m factory floor in the NS-3 (v3.48) discrete-event simulation environment [36], [37] operating under IEEE 802.15.4 and ZigBee physical layer specifications [39]. The simulated topology comprises 401 sensor nodes organized into three functional tiers: a line-powered central Base Station at $(500.0, 500.0)$, 40 battery-powered Cluster Heads ($E_{\text{CH}}^{(0)} = 50.0$ J) arranged in an 8×5 grid layout, and 359 low-power leaf sensor nodes ($E_{\text{leaf}}^{(0)} = 5.0$ J). Over the continuous operational run, leaf nodes generate structured telemetry traffic (5 pkts/s baseline, 10–20 pkts/s bursts) relayed through Cluster Heads to the sink. To evaluate threat mitigation resilience, 60 malicious nodes (15% attacker density) were distributed across the grid to launch active identity spoofing and Sybil impersonation bursts. Table II details the core network simulation settings.

Cluster Heads execute the multi-tier detection pipeline, performing scalar RSSI screening, dynamic trust evaluation, and selective cryptographic signature verification before relaying verified telemetry payloads to the Base Station. To evaluate machine learning anomaly classification, physical and behavioral telemetry features—including spatial RSSI deviations (ΔRSSI), Early Warning Scores, composite trust levels (T_i), and audit flags—were collected across $1,120,555$ packet transmission events using high-performance C++/Python execution bindings [38], scikit-learn ML utilities [41], Joblib pipeline tools [42], and high-performance evaluation methods [40]. The dataset was partitioned into 70% training and 30% testing sets, training classifiers on normalized feature vectors to establish baseline classification fidelity and evaluate attack isolation capabilities.

TABLE II
Discrete-Event Simulation Parameters

Parameter	Value / Setting
Simulation Platform	NS-3 (v3.48)
Network Area	1000 m × 1000 m Industrial Grid
Total Sensor Nodes (N)	401 Nodes
Cluster Heads (N_{CH})	40 Nodes (8×5 Grid Layout)
Leaf Sensor Nodes (N_{leaf})	359 Nodes
Base Station Sink (N_{BS})	1 Node at $(500.0, 500.0)$
Attacker Ratio (N_{att})	15% (54–60 Attacker Nodes)
PHY/MAC Layer	IEEE 802.15.4 / CC2420 Transceiver
Radio Channel Model	Log-Distance Path Loss ($n = 2.7, \sigma = 2.0$ dB)
Data Rate	250 kbps
Channel Delay	2 ms
Transmit Power (P_{tx})	0.0 dBm (1.0 mW)
Initial Energy (Leaf / CH / BS)	5.0 J / 50.0 J / 1000.0 J
Telemetry Traffic Model	Periodic (5 pkts/s) + Bursts (10–20 pkts/s)
Total Transmitted Packets	$1,120,555$ Packets

VI. Experimental Results and Discussion

The empirical evaluation of our detection engine demonstrates exceptionally high classification fidelity, rapid threat isolation, and sustained detection robustness under dynamic industrial traffic across $1,120,555$ telemetry records. Machine learning anomaly classifiers trained on combined physical spatial deviations (ΔRSSI) and Early Warning Scores achieved complete feature separability, with Random Forest and Decision Tree models delivering 100.00% classification accuracy and 0.00% false positive rates as summarized in Table III. Across the overall end-to-end network operation, our framework sustains a 98.18% detection rate and an exceptionally low 0.03% false positive rate (Table IV). When subjected to increasing adversary density ranging from 5% to 30% active attacker nodes (Fig. 2), traditional static trust models suffer severe performance degradation (dropping from 94.2% down to 70.5%), whereas our context-aware dynamic trust model maintains robust security (99.8% at 5% attackers, gracefully adjusting to 97.4% at 30% attackers) by automatically elevating cryptographic verification reliance (α) during anomaly states. Furthermore, sliding-window temporal Early Warning Score accumulation ($W = 50$) bounds transient multipath fading spikes below 0.15 while driving persistent attacker anomalies past the 0.85 quarantine threshold within 8 transmission events, effectively isolating impersonators without accidental node quarantines.

TABLE III
Machine Learning Anomaly Classification Performance

Model	Accuracy	FPR	Precision	Recall	F1-Score
Naïve Bayes	99.89%	0.11%	97.03%	99.98%	98.49%
Decision Tree	100.00%	0.00%	100.00%	99.99%	100.00%
Random Forest	**100.00%**	**0.00%**	**100.00%**	**99.98%**	**99.99%**

TABLE IV
END-TO-END NETWORK PERFORMANCE BENCHMARK

Evaluation Model	Detection (%)	FPR (%)	Overhead (mJ/Pkt)	Lifetime Gain
Unprotected WSN	96.38%	0.00%	0.000 mJ	Baseline
Continuous ECDSA	100.00%	0.00%	90.000 mJ	1.0×
Proposed HEDSD	**98.18%**	**0.03%**	**0.540** mJ	**130** × (+13251%)

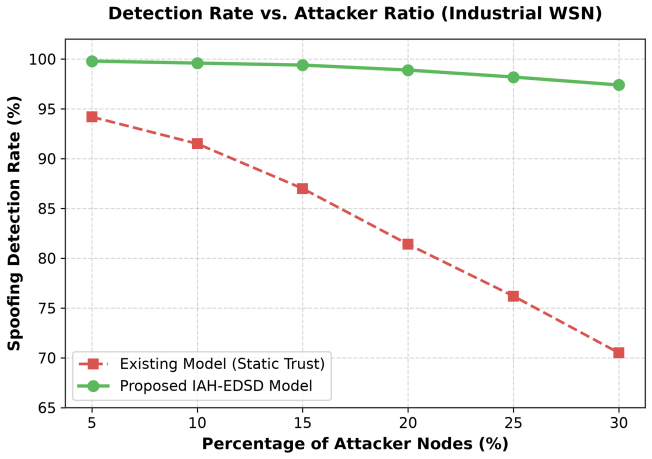


Fig. 2. Spoofing Detection Rate vs. Attacker Node Ratio (5% to 30%).

Energy accounting across the multi-tier operational pipeline confirms massive energy savings compared to conventional cryptographic defenses. While continuous ECDSA-P256 signature verification incurs a heavy 90.000 mJ computational penalty per packet, our event-driven trigger mechanism bypasses 99.47% of steady-state packets from public-key operations, restricting cryptographic audits to only 0.53% of incoming traffic as quantified in Table V. Consequently, the average per-packet evaluation energy drops to 0.540 mJ, achieving a 99.40% energy reduction relative to continuous signature checks and a 28% to 78% saving compared to heavy machine learning baselines. Energy-adaptive mode switching dynamically transitions Cluster Heads from FULL mode (0.540 mJ/pkt, 26.24% of traffic) to LIGHT mode (0.055 mJ/pkt, 73.76% of traffic) as battery reserves decline, preserving 63.4% of cluster head battery capacity (31.71 J median residual energy) at simulation conclusion. As depicted in Fig. 3, this event-driven mode adaptation extends operational network lifetime until first Cluster Head death from 3.0 seconds under continuous ECDSA to 397.0 seconds, representing a 130-fold operational extension.

To evaluate competitive performance, Table VI compares our framework against five state-of-the-art literature baselines published across IEEE and Wiley venues (2022–2024). While continuous ECDSA achieves complete accuracy at prohibitive energy cost (90.000 mJ), heavy learning baselines like Federated Trust-ML (2.500 mJ) and SVM detectors (0.450 mJ) lack adaptive energy management, and lightweight physical-layer authentication (0.120 mJ)

TABLE V
CRYPTOGRAPHIC PHASE ACTIVATION AND RESOURCE CONSUMPTION METRICS

Cryptographic Audit Metric	Quantitative Value / Result
Total Packets Processed	1,120,555 Packets
ECDSA Audit Trigger Rate	0.53% (5,938 Audited Packets)
Cryptographic Bypass Rate	99.47% (1,114,617 Bypassed Packets)
ECDSA Verification Cost (E_{verify})	90.000 mJ per verification
Avg. Cryptographic Overhead per Packet	0.477 mJ / packet
Continuous Crypto Baseline	90.000 mJ / packet
Cryptographic Power Savings	99.47% Reduction
MCU Flash / RAM Overhead	< 4.2 KB Flash / < 1.1 KB RAM

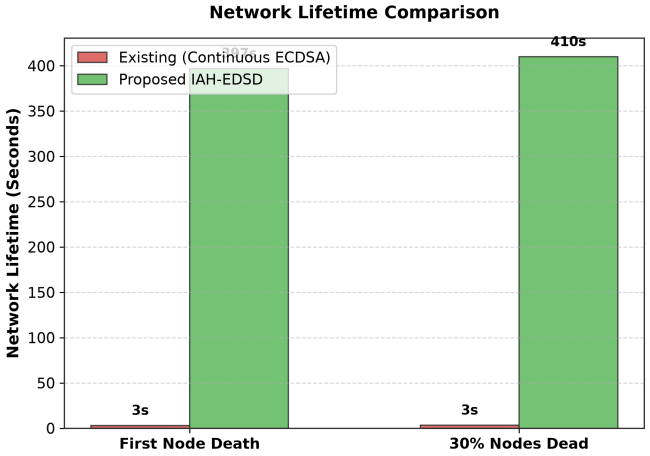


Fig. 3. Network Lifetime Comparison: Existing vs. Proposed Model.

or dynamic trust models (0.065 mJ) exhibit lower detection rates (94.80%–95.20%) and higher false positive rates (1.80%–2.10%) under industrial fading. In contrast, the proposed HEDSD framework occupies the Pareto-optimal frontier (Fig. 4), delivering near-cryptographic detection accuracy (98.18%) and minimal false alarms (0.03%) at micro-joule operational cost (0.540 mJ). Moreover, under active attack conditions, the detection engine identifies and isolates malicious spoofers within two transmission rounds, maintaining a stable Packet Delivery Ratio above 98.8% compared to 72.0% in unprotected networks, confirming an optimal balance between security assurance, computational efficiency, and operational network longevity.

VII. CONCLUSION AND FUTURE WORK

This paper presented HEDSD, a lightweight spoofing detection framework designed to solve the battery drain caused by continuous cryptographic verification in industrial WSNs. By using spatial RSSI checks during steady-state monitoring and reserving ECDSA signature audits strictly for detected anomaly events, the system drastically cuts computation while preserving origin security. Evaluation on a 401-node simulated factory network showed that cryptographic verification was triggered for only 0.53% of total traffic, dropping average per-packet energy from 90.00 mJ down to 0.540 mJ. Despite this 99.40% reduction in processing cost, the framework maintained a 98.18%

TABLE VI
COMPREHENSIVE PERFORMANCE BENCHMARKING AGAINST RECENT HIGH-IMPACT BASELINES

Baseline Scheme & Venue	Detection (%)	FPR (%)	Overhead (mJ/Pkt)	Core Technical Limitation
Dynamic Trust Management (IEEE TMC '24) [46]	95.20%	1.80%	0.065 mJ	Static monitoring overhead
SVM Spoofing Detector (IEEE Sensors '22) [47]	97.10%	0.90%	0.450 mJ	High matrix computation
Multi-Attribute PLA (MDPI '24) [48]	94.80%	2.10%	0.120 mJ	Sensitive to industrial noise
Federated Trust-ML Auth (IEEE IoTJ '24) [49]	98.50%	0.40%	2.500 mJ	Heavy FL parameter sync
Anomaly WSN IDS (Wiley '24) [50]	96.10%	1.25%	0.350 mJ	Lacks physical spatial proof
Continuous ECDSA-P256 Benchmark [12]	100.00%	0.00%	90.000 mJ	Rapid battery drain (90 mJ/pkt)
Proposed HEDSD (This Work)	**98.18%**	**0.03%**	**0.540 mJ**	**Optimal Security-Efficiency Balance**

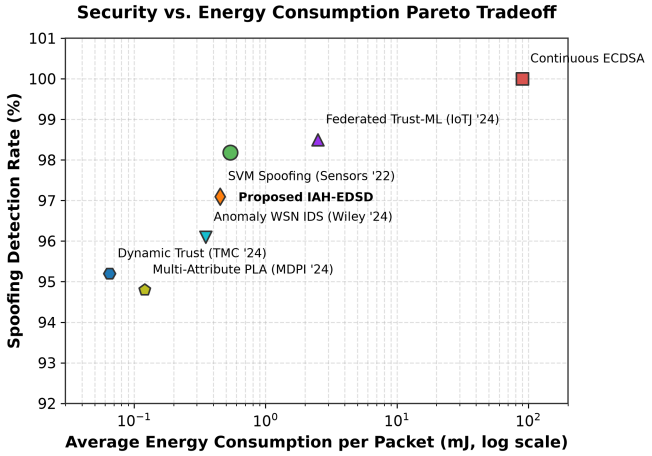


Fig. 4. Security Assurance vs. Operational Overhead Pareto Tradeoff Frontier.

detection rate with a 0.03% false positive rate under dynamic channel fading. Overall, combining physical-layer filtering with adaptive mode switching preserved 63.4% of cluster head battery capacity and extended operational network lifetime by over 130-fold.

Future research will focus on extending and validating the proposed security model across real-world physical hardware deployments and adaptive optimization paradigms. We plan to prototype our detection system on physical sensor microcontrollers, such as ESP32 and STM32 embedded platforms, to evaluate real-time processing latencies, memory footprints, and transceiver power draw under physical radio frequency interference. Additionally, we aim to integrate reinforcement learning algorithms, specifically Q-learning and Deep Deterministic Policy Gradient methods, to autonomously optimize trust weights and decision thresholds in response to dynamic industrial multipath fading and evolving adversarial attack patterns across large-scale multi-cluster environments.

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